



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Aerospace Supplier Risk Intelligence

The domain of the Project: Data Analytics

G1-DA

Team Mentors (and their designation):

Nithyasri Kannathal – Software Development Engineer II at Predigle

Team Members:

Ms. Simranjot Kaur

Period of the project

February 2026 to July 2026



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Declaration

The project titled “Aerospace Supplier Risk Intelligence” has been mentored by Ms.Nithyasri Kannathal, organised by SURE Trust, from April 2023 to August 2023, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. Simranjot Kaur

Signature: Simranjot Kaur

Mentor’s Name

Ms.Nithyasri Kannathal

Software Development Engineer II

Predigle

Prof. Radhakumari

Executive Director & Founder

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Executive Summary

The **Aerospace Supplier Risk Intelligence** project is a data analytics and machine-learning solution developed to identify purchase orders that may be delivered late.

Procurement teams handle many suppliers, parts and purchase orders. Checking every order manually takes time and can result in important risks being identified too late. This project helps users focus on the purchase orders that require attention first.

The project uses data related to:

- 40 suppliers
- 300 parts
- 29,666 purchase orders
- 368 quality incidents
- 280,800 supply-chain history records
- 6,002 final PO prediction records

Three machine-learning models were compared: Logistic Regression, Random Forest and HistGradientBoosting. A classification threshold of **0.45** was used for the early-warning system. At this threshold, the model achieved **63.9% precision, 71.9% recall and a 67.7% F1 score**.

The project generated 6,002 purchase-order alerts, including:

- 285 Critical alerts
- 2,410 High alerts
- 1,136 Medium alerts
- 2,171 Low alerts

The final results are presented through a Streamlit website and Power BI dashboards. Streamlit helps users investigate individual purchase orders, understand the main risk driver and view recommended actions. Power BI provides summary charts and management-level reporting.

The current solution is a proof-of-concept. It supports procurement decisions but does not make automatic decisions or guarantee that an order will be late.



Introduction

2.1 Background

Aerospace companies depend on suppliers for the timely delivery of important parts and materials. A delay in one critical part can affect production schedules, inventory availability and customer commitments.

Procurement teams may need to monitor thousands of purchase orders. Manually reviewing every order is difficult and time-consuming. Therefore, an early-warning system can help users identify possible delays before they create a larger operational problem.

2.2 Problem Statement

The main problem is that purchase-order risks are often identified too late.

The major business challenges are:

- Manual monitoring of thousands of purchase orders
- Difficulty in identifying high-risk orders
- Lack of clear alert priorities
- Limited explanation behind a risk score
- Difficulty in deciding what action should be taken

This project addresses these challenges by predicting PO late-delivery risk and creating a prioritised action queue.

2.3 Scope

The project covers:

- Data cleaning and validation
- Feature engineering
- SQL-based analysis
- Machine-learning model comparison
- PO late-delivery probability
- Risk categorisation
- Rule-based risk explanations
- Recommended procurement actions
- Streamlit application development
- Power BI dashboard development



2.4 Limitations

The main limitations are:

- The project is a proof-of-concept.
- It is not connected to a live ERP system.
- The Streamlit application uses prepared project output files.
- The model has moderate predictive performance.
- The explanation system is rule-based and does not use an external LLM.
- The model does not automatically approve, reject or cancel orders.
- Human review is required before taking procurement action.
- Email, Teams and n8n notifications are not currently implemented.



Project Objectives

The main objective of this project is to develop an early-warning system for purchase-order delivery risk.

The specific objectives are:

1. To combine supplier, part, purchase-order, quality and historical supply-chain data.
2. To clean and validate the project datasets.
3. To create useful features related to supplier performance and delivery risk.
4. To compare different machine-learning classification models.
5. To estimate the late-delivery probability of individual purchase orders.
6. To classify alerts into Critical, High, Medium and Low categories.
7. To identify the main risk driver behind each alert.
8. To provide simple explanations and recommended procurement actions.
9. To develop an interactive Streamlit application.
10. To create Power BI dashboards for visual analysis and management reporting.
11. To provide a prioritised and downloadable procurement action queue.



Methodology and Results

4.1 Tools and Technologies Used

The following tools and technologies were used:

Tool	Purpose
Python	Data cleaning, analysis and machine learning
Pandas and NumPy	Data processing
Scikit-learn	Model training and evaluation
SQL	Analytical queries and reusable views
Streamlit	Interactive web application
Plotly	Interactive charts
Power BI	Dashboards and management reporting
GitHub	Project code and documentation

4.2 Dataset Description

The project uses the following datasets:

Dataset	Records	Purpose
Suppliers	40	Supplier information
Parts Master	300	Part details and criticality
Purchase Orders	29,666	Order and delivery information
Quality Incidents	368	Supplier and part quality issues



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Dataset	Records	Purpose
Supply-Chain History	280,800	Historical delivery information
PO Predictions	6,002	Model prediction output
Final PO Alerts	6,002	Priority, reason and action output

4.3 Data Cleaning and Validation

Before model development, the datasets were checked and prepared.

The main activities included:

- Checking missing values
- Checking duplicate records
- Correcting data types
- Standardising categorical values
- Checking supplier and part identifiers
- Reviewing date fields
- Confirming record counts
- Preparing clean data for analysis

Validation Results

Validation Check	Expected Result	Status
Suppliers	40	Passed
Parts	300	Passed
Purchase orders	29,666	Passed
Quality incidents	368	Passed
PO predictions	6,002	Passed



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Validation Check	Expected Result	Status
Final PO alerts	6,002	Passed
Critical alerts	285	Passed
Critical-part escalations	152	Passed

All important record-count checks were passed.

4.4 Feature Engineering

Feature engineering was performed to create useful inputs for the machine-learning models.

Important features included:

- Supplier historical late-delivery rate
- Promised lead days
- Lead-time days
- Part criticality
- Delivery risk
- Quality risk
- Compliance risk
- Operational risk

The three most important features were:

Feature	Importance	Meaning
Supplier Historical Late Rate	29.45%	Previous supplier delivery reliability
Promised Lead Days	13.45%	Committed delivery period
Lead Time Days	12.61%	Time required to fulfil an order

Supplier Historical Late Rate was the most important feature. This means that a supplier's previous delivery performance is a useful signal for identifying current PO risk.



4.5 SQL Analytics

Nine analytical SQL views were created:

1. vw_SupplierRisk
2. vw_SupplierRiskSummary
3. vw_CriticalSupplierQueue
4. vw_POPredictions
5. vw_POAlertQueue
6. vw_InventoryHealth
7. vw_QualityIncidents
8. vw_DeliveryPerformance
9. vw_ExecutiveKPIs

These views organise the cleaned data into reusable outputs for analysis, reporting and dashboard development.

4.6 Machine Learning

The project was treated as a binary-classification problem. The model predicts whether a purchase order is likely to be late.

Three models were compared:

- Logistic Regression
- Random Forest
- HistGradientBoosting

Model Comparison

Model	Accuracy	ROC-AUC
Logistic Regression	63.6%	64.3%
Random Forest	63.3%	64.6%
HistGradientBoosting	63.2%	64.2%

Random Forest had the highest ROC-AUC, while Logistic Regression had the highest accuracy. The results of all three models were close.



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The model is used as an early-warning tool. It does not guarantee that a purchase order will be late.

4.7 Threshold Results

A threshold of **0.45** was used to identify purchase orders requiring attention.

Metric	Result
Threshold	0.45
Precision	63.9%
Recall	71.9%
F1 Score	67.7%
Alert Rate	63.8%
ROC-AUC	Approximately 0.64

Recall was important because the aim was to identify as many possible late-risk purchase orders as reasonably possible.

A recall of 71.9% means that the model identified approximately 71.9% of the late-risk cases in the evaluated data.

4.8 Risk Analysis

The 6,002 prediction records were divided into four alert categories.

Alert Priority	Number of Alerts	Percentage
Critical	285	4.7%
High	2,410	40.2%
Medium	1,136	18.9%
Low	2,171	36.2%
Total	6,002	100%



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Important results include:

- 2,695 Critical and High alerts
- 285 Critical alerts
- 152 critical-part escalations
- 6,002 total PO alerts

Alerts are prioritised in the following order:

Critical → High → Medium → Low

Within each category, orders with a higher late-delivery probability are displayed first.

4.9 Explainable Risk Recommendations

The project includes a rule-based explanation system.

The explanation workflow is:

Risk Probability → Primary Risk Driver → Explanation → Recommended Action

The system explains the main reason behind an alert and suggests an appropriate procurement action.

Example actions include:

- Contact the supplier.
- Confirm the delivery date.
- Request a recovery plan.
- Increase monitoring.
- Review alternate suppliers.
- Escalate a critical-part order.

The explanation system is rule-based. It does not use ChatGPT or another external Large Language Model.

4.10 Streamlit Application

The Streamlit website is the interactive decision-support part of the project.

It contains the following pages:

- Project Overview
- Data and SQL Pipeline
- Machine Learning



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- Explainable Risk Recommendations
- Procurement Decision Center
- Power BI Dashboard Showcase

The website allows users to:

- Review project KPIs
- Compare machine-learning models
- Select suppliers and purchase orders
- View late-delivery probability
- Identify the primary risk driver
- Read the risk explanation
- View the recommended action
- Filter alerts
- Download the procurement action queue

The Streamlit application supports detailed PO-level investigation.

4.11 Power BI Dashboards

Five Power BI dashboard pages were created:

1. Executive Control Tower
2. Supplier Risk Analysis
3. Predictive Delivery Risk
4. Quality and Inventory Risk
5. Procurement Action Queue

Power BI provides summary charts, KPIs and management-level analysis.

The difference between Streamlit and Power BI is:

- **Streamlit** supports detailed PO investigation, explanation and action.
- **Power BI** provides overall visual analysis and reporting.

4.12 Project Architecture

The project follows this workflow:

CopyRaw Data





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Data Cleaning



Data Validation



Feature Engineering



Machine-Learning Prediction



Risk Categorisation



Risk Explanation



Recommended Action



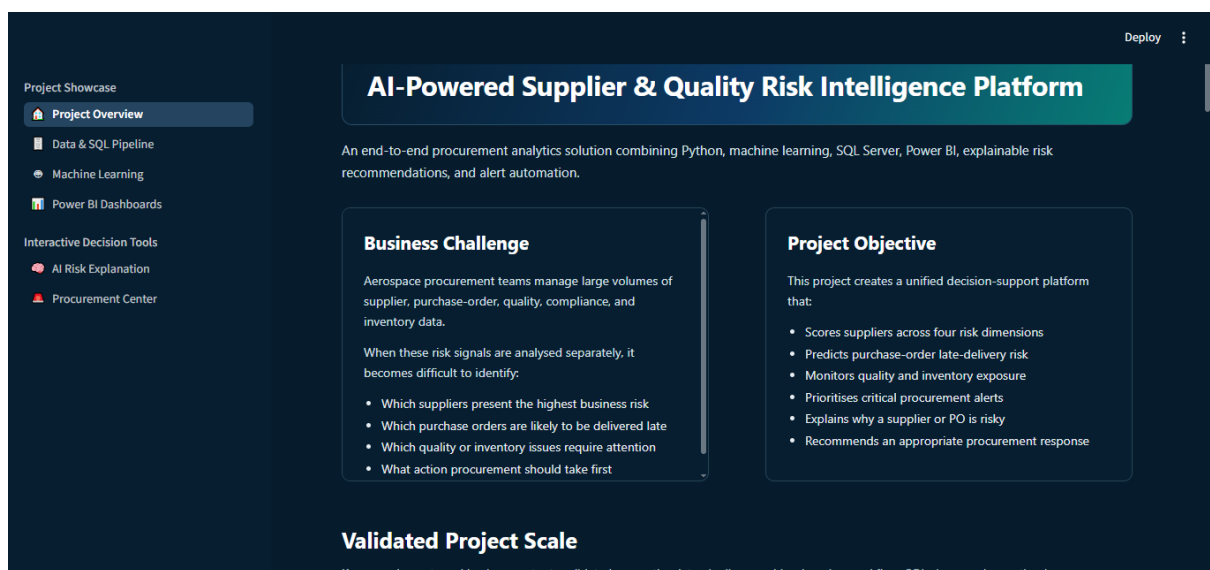
Streamlit Application and Power BI Dashboards

This architecture connects raw data with practical procurement decisions.

4.13 Working Screenshots

Insert the following screenshots in this section:

1: Project Overview

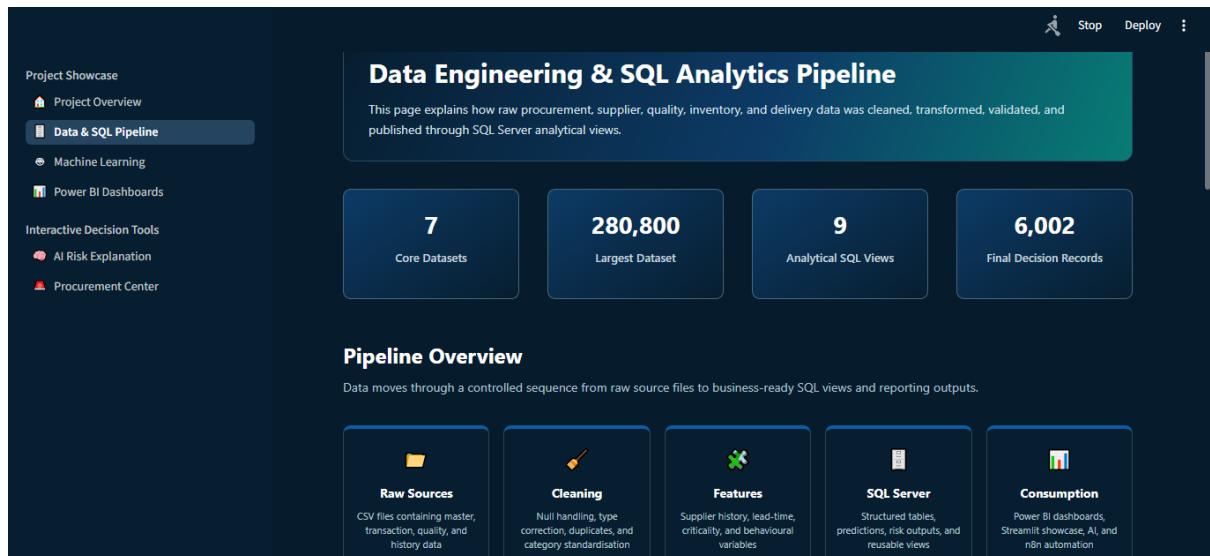




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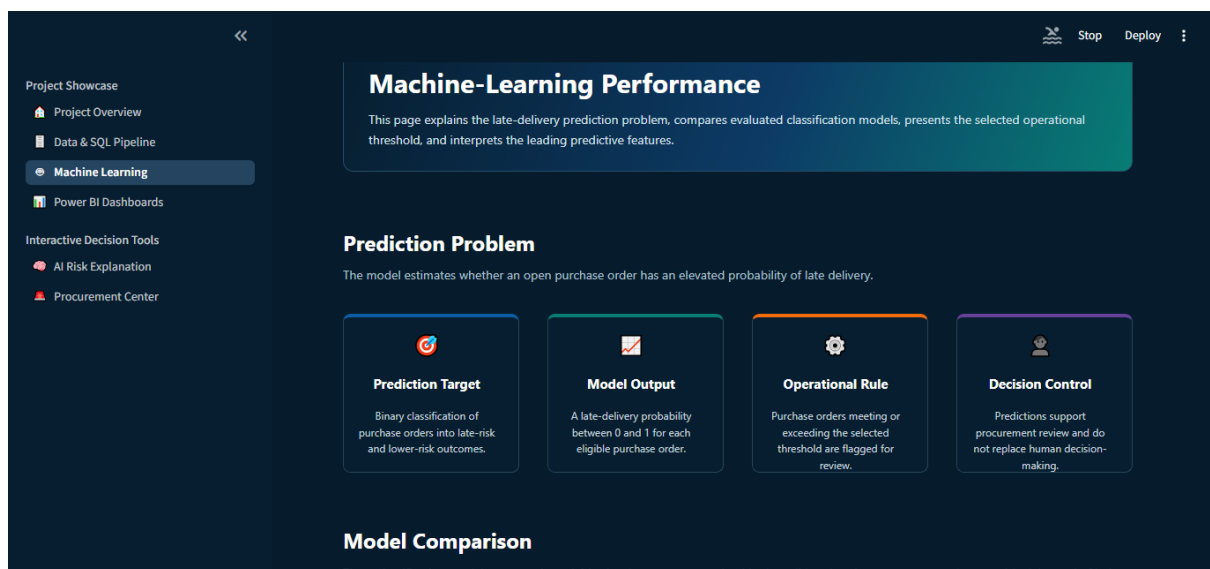
This page displays the project summary, KPIs and architecture.

2: Data and SQL Pipeline



This page shows datasets, cleaning steps, validation checks and SQL views.

3: Machine Learning



This page shows model comparison, threshold metrics and feature importance.

4: Explainable Risk Recommendations



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AI Supplier Risk Explanation

Select a supplier to review multidimensional risk scores, purchase-order alerts, leading risk drivers, and a grounded business explanation generated only from available project data.

Grounded explanation mode: This page does not invent supplier facts. Explanations are created from the selected supplier's risk scores and associated PO alerts. No external AI API key is required for this version.

Select Supplier

Choose a supplier to generate an interactive risk profile and procurement explanation.

Supplier: SUP001 | Explanation format: Executive Summary

Supplier SUP001 — Low Risk

Review the supplier's overall score, individual risk dimensions, purchase-order alerts, and recommended procurement response.

This page displays the risk probability, primary risk driver, explanation and recommended action.

5: Procurement Decision Center

Procurement Decision Center

Filter, prioritise, investigate, and export purchase-order alerts using late-delivery probability, supplier risk, part criticality, alert priority, and primary risk driver.

Action Queue Filters

Leave a filter empty to include all available values.

Filters are applied together. For example, selecting Critical priority and Criticality A displays only records meeting both conditions.

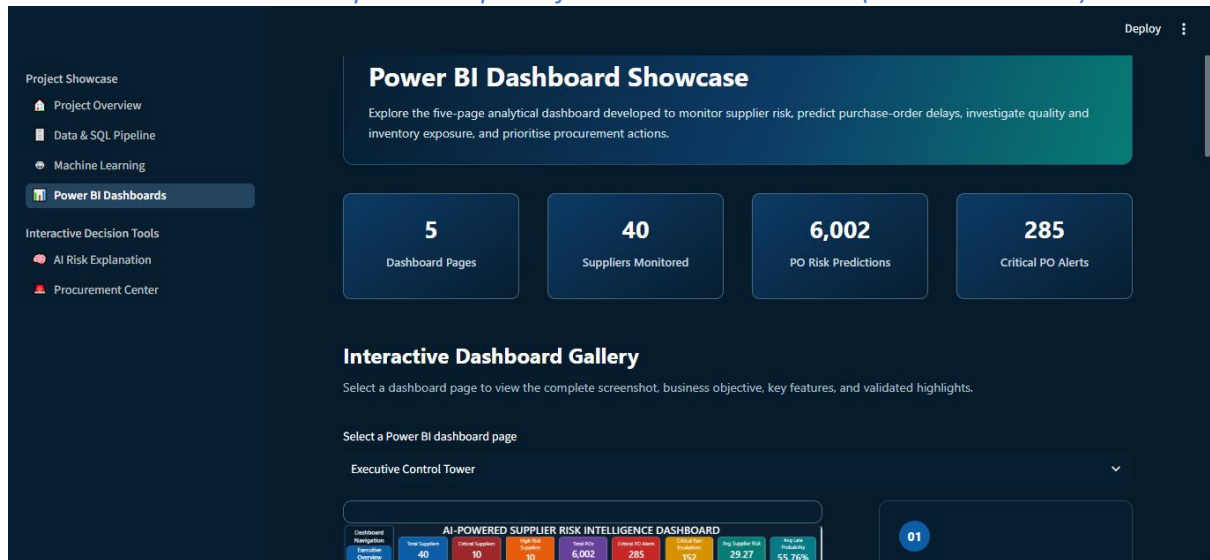
Alert Priority: Choose options | Risk Category: Choose options | Supplier: Choose options | Criticality: Choose options | Primary Risk Driver: Choose options | Clear Filters

This page allows users to filter alerts and review individual purchase orders.

6: Power BI Dashboard



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The Power BI dashboards provide summary visualisations and procurement KPIs.

4.14 GitHub Link

The project source code, SQL files and documentation are available at:

GitHub Repository Link: <https://github.com/simranjot21/aerospace-supplier-risk-intelligence>

Suggested repository name:

aerospace-supplier-risk-intelligence



Learning and Reflection

This project helped me understand how a business problem can be converted into a complete data analytics and machine-learning solution.

I learned how to:

- Understand a procurement business problem
- Work with multiple datasets
- Clean and validate data
- Create useful analytical features
- Develop SQL views
- Compare machine-learning models
- Understand precision, recall, F1 score and ROC-AUC
- Create an interactive Streamlit application
- Design Power BI dashboards
- Explain technical results in simple business language
- Create a prioritised action queue

The most important learning was that a prediction alone is not enough. Users also need to understand why a purchase order is risky and what action they should take.

I also learned that machine-learning models are not always perfect. The results must be explained honestly, and final decisions should remain with the user.

This project improved my technical knowledge, problem-solving ability, communication skills and confidence in working on an end-to-end data analytics project.



Conclusion and Future Scope

7.1 Future Scope

The project can be improved in the future by adding:

- Live ERP and SQL integration
- Automatic scoring of new purchase orders
- Email or Microsoft Teams notifications
- Optional workflow automation using n8n
- Buyer feedback collection
- Model monitoring and retraining
- Role-based access
- Supplier capacity and logistics data
- Production deployment with secure authentication

Notification automation and n8n were not implemented in the current project. They are included only as future improvements.

7.2 Conclusion

The Aerospace Supplier Risk Intelligence project was developed to identify purchase orders that may be delivered late.

The project successfully combined data cleaning, SQL analytics, machine learning, risk explanation, Streamlit and Power BI into one decision-support workflow.

It generated 6,002 PO alerts, including 285 Critical alerts and 152 critical-part escalations. The selected threshold achieved 71.9% recall, 63.9% precision and a 67.7% F1 score.

The Streamlit application helps users investigate individual purchase orders, understand the primary risk driver and review recommended actions. Power BI provides overall visual analysis and management reporting.

The project is currently a proof-of-concept. It does not make automatic procurement decisions and does not guarantee that a purchase order will be late. However, it shows how data analytics and machine learning can help procurement teams move from manual and reactive monitoring towards a more proactive and organised approach.



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